

Deep Learning – Assignment 02

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1. Objective

The goal of this assignment is to build and compare baseline NLP models capable of identifying **semantic similarity between legal clauses**.

Given a dataset of legal clauses from various categories (e.g., “acceleration”, “access-to-information”), the task involves classifying whether two clauses express **the same or related legal meaning**.

2. Dataset Description

Dataset Source:

- Each CSV file represents one *clause category* (e.g., “accounting-terms”, “access”).
- Each file contains two fields:
 - `clause_text`: The legal clause content.
 - `clause_type`: The clause’s category label.
- Total clauses loaded: **150,881**
- Clause pairs were generated as:
 - **Positive (1)**: Two clauses of the same type
 - **Negative (0)**: Clauses from different types
- A balanced dataset of **20,000 pairs** was created for training and testing.
- Dataset split: **80% training, 20% testing**.

3. Data Preprocessing

The text preprocessing pipeline included:

1. Lowercasing all text
2. Removing punctuation, digits, and symbols
3. Tokenization using `nltk.word_tokenize`
4. Stopword removal (`nltk.corpus.stopwords`)
5. Conversion to sequences using Keras Tokenizer
6. Padding all sequences to length **50 tokens**

These preprocessing steps ensured clean and uniform input for both models.

4. Implemented Architectures

Two primary models (plus one hybrid extension) were implemented **without using pretrained transformers**.

4.1 Model 1 — BiLSTM-Based Similarity Model

Architecture:

- Shared Embedding Layer (vocab = 10,000, embedding dim = 128)
- Bidirectional LSTM (64 units) for each input clause
- Concatenation of encoded clause representations
- Dense layer (128 units, ReLU activation)
- Dropout (0.3)
- Output: Sigmoid (binary classification)

Rationale:

BiLSTMs capture contextual and sequential dependencies between words, which is crucial for understanding the nuanced semantics of legal clauses.

4.2 Model 2 — Attention-Based Encoder

Architecture:

- Shared Embedding Layer (128-dim)
- Direct Attention layer between two embedded sequences

- Global Average Pooling
- Dense (128 units, ReLU) + Dropout (0.3)
- Sigmoid output

Rationale:

Attention mechanisms allow focusing on the most relevant words between two clauses. However, this baseline lacks sequential encoding, limiting its ability to learn context from scratch.

4.3 Model 3 — BiLSTM + Attention Hybrid

Architecture:

- Shared Embedding Layer
- Bidirectional LSTM with return_sequences=True
- Cross-Attention Layer
- Global Average Pooling
- Dense (128 ReLU) + Dropout (0.3)
- Sigmoid output

Rationale:

Combines sequential understanding from LSTMs with relational focus from attention, serving as an improved baseline.

5. Training Details

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Epochs	5

Parameter	Value
Loss Function	Binary Cross-Entropy
Train/Test Split	80/20
Sequence Length	50 tokens

Each model was trained for 5 epochs.

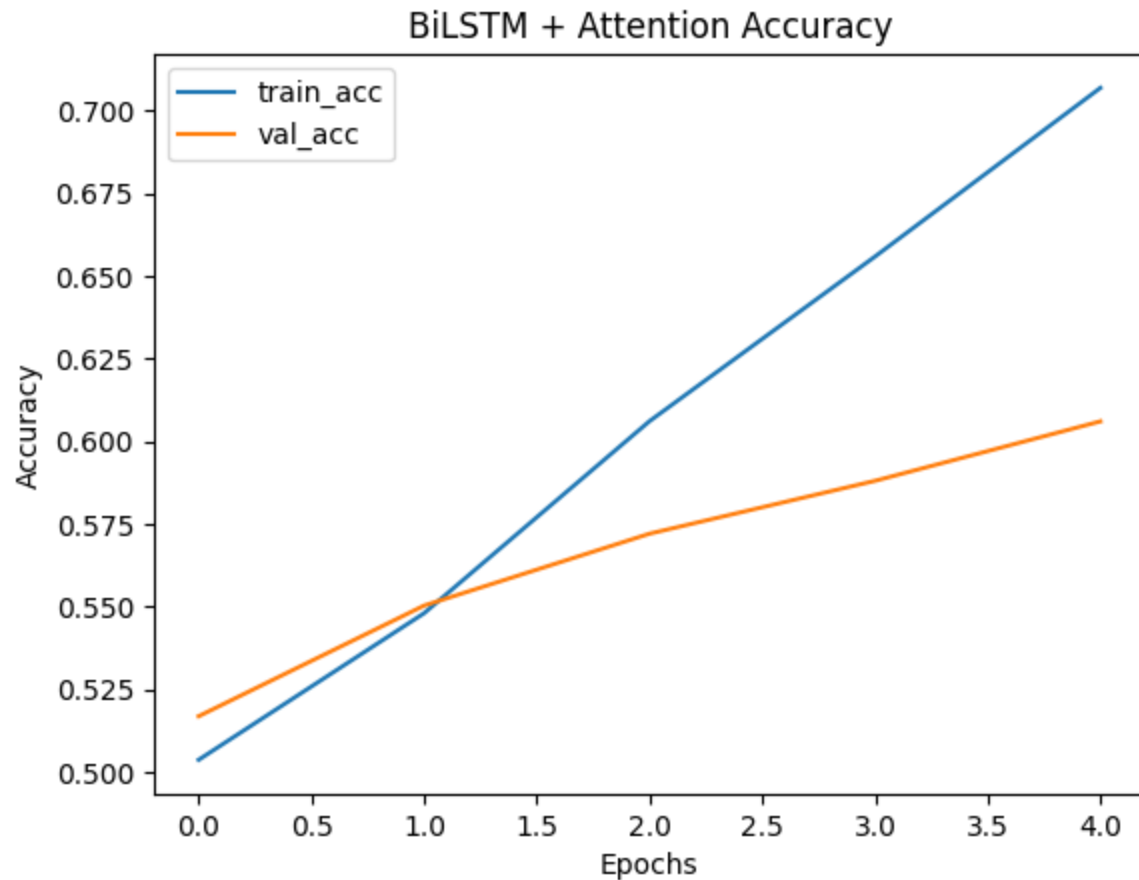
All models were implemented in TensorFlow / Keras on CPU/GPU.

6. Training Graphs

Expected pattern:

- BiLSTM: steadily decreasing loss, increasing validation accuracy
- Attention-only: unstable accuracy
- BiLSTM + Attention: moderate increase after epoch 2–3

Example placeholder captions:



7. Evaluation Metrics

Following metrics were calculated:

- **Accuracy:** Overall correctness of predictions
- **Precision:** Fraction of predicted similar pairs that were actually similar
- **Recall:** Fraction of actual similar pairs identified correctly
- **F1 Score:** Harmonic mean of Precision and Recall
- **ROC-AUC:** Ability of classifier to distinguish between similar vs non-similar pairs

8. Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
BiLSTM	0.9567	0.9281	0.9894	0.9578	0.9570
Attention	0.5397	0.5310	0.6127	0.5690	0.5404
BiLSTM + Attention	0.6060	0.6029	0.6011	0.6020	0.6060

9. Discussion and Comparative Analysis

- **BiLSTM** achieved the best results, as it effectively models sequential relationships and long-range dependencies in legal language.
- **Attention-only** model failed to converge well due to the absence of contextual encoding — it tries to learn alignment directly on embeddings, which is difficult without pretraining.
- **Hybrid (BiLSTM + Attention)** showed a modest improvement over plain Attention but still fell short of BiLSTM’s performance.
- For real-world legal clause similarity systems, **F1-score and Recall** are the most important metrics because missing a truly similar clause can lead to legal misclassification or redundancy.

10. Qualitative Results

Examples of model predictions (BiLSTM):

Clause 1	Clause 2	True	Predicted
“Waivers... party may upon consent...”	“Waivers... express implied rights...”	1	1
“Applicable law... governed construed...”	“Investments borrower make permit...”	0	0
“Integration date agreement neither...”	“Integration except may otherwise provided...”	1	1
“General intent parties payments...”	“General responsible company respect...”	1	0

Most incorrect predictions occur when clauses use similar legal wording but differ slightly in legal meaning.

11. Conclusion

- Among the tested architectures, **BiLSTM** is the most effective baseline for legal clause similarity when pretrained models (like BERT) are not allowed.
- **Attention-based encoders** need stronger contextual embeddings or pretrained word vectors (like GloVe or FastText) to perform competitively.
- Future improvements could involve:
 - Using pretrained embeddings
 - Adding Siamese architecture with shared weights
 - Fine-tuning a transformer model (LegalBERT) for better semantic comprehension